Akash Kundu

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Portfolio | GitHub | LinkedIn

Summary

AI/ML & Systems Engineer with core competence in **Computer Vision**, **Medical Informatics**, **Edge Model Acceleration (ONNX/FP16)**, and **High-Performance Distributed Systems**. Lead author of **OphthalmoAI** (calibrated tri-backbone vision ensemble, 85.18% accuracy, 0.9818 Macro AUROC, 2.15x serving acceleration) with peer-reviewed research manuscript in preparation. Experienced in building sovereign/local AI pipelines (Whisper Bengali speech processing, local Ollama fallback) and secure microservices. Strong academic foundation with **SGPA 7.93** in latest completed semester, current **CGPA 7.59**, and **95.4% in Class X (CBSE)**.

Education (Class X Onwards)

Techno India University Sep 2023 -- Sep 2027 (Expected)
B.Tech in Computer Science and Engineering; ***CGPA: 7.59*** *(75.9%)* *Kolkata, West Bengal, India*

- **Latest Completed Semesters: SGPA: 7.93** (5th Semester) | **SGPA: 8.29** (4th Semester).

Arambagh Vivekananda Academy (CBSE) April 2021 -- May 2023
Higher Secondary Certificate (Class XII Science); ***Percentage: 71.6%*** *Arambagh, West Bengal, India*

Arambagh Vivekananda Academy (CBSE) April 2020 -- May 2021
Secondary School Examination (Class X); ***Percentage: 95.4%*** *Arambagh, West Bengal, India*

Professional Experience

Purposive AI July 2026 -- Present
Software Engineering Intern (Backend & Systems) *Kolkata, India (Hybrid)*

- Architected an asynchronous ledger pipeline using FastAPI, Celery, and Whisper, translating conversational **Bengali speech audio** into bank-submittable structured records and cutting manual bookkeeping effort by **80%**.
- Engineered an automated fallback circuit breaker to local **Ollama LLM** instances when cloud APIs degraded, ensuring sovereign data governance, 99.9% uptime, and slashing operational inference expenditures by **87%**.
- Designed high-throughput task worker queues with PostgreSQL **pgvector** semantic state tracking, sustaining **sub-500ms** turnaround under production SLAs; enforced 100% TDD pass rate across 156 Pytest test suites.
- Implemented token-bucket rate limiting, structured transactional logging, and strict input sanitization to guarantee complete auditability across concurrent user transactions.

Projects & Research

OphthalmoAI -- Point-of-Care Retinal Disease Screening Platform | *PyTorch, ONNX, FastAPI, React 19* GitHub

- Architected a calibrated soft-voting ensemble (**DenseNet-201, ConvNeXt-Small, EfficientNet-V2-M**) with Platt temperature scaling (**0.0381 ECE**), achieving **85.18% accuracy and 0.9818 Macro AUROC** over 6 retinal classes ($n=938$).
- Engineered pre-inference biophysical domain guardrails (**OAC-DG**) evaluating circular aperture geometry and chromophore backscatter ($\rho_{\text{RB}} \ge 1.05$), deterministically blocking 100% non-fundus imagery before GPU allocation.
- Validated across external clinical cohorts: **91.30% DR recall, 100% Proliferative DR recall** on IDRiD (India), and **100% fail-safe clinical escalation** on out-of-distribution optical crops on RIM-ONE DL (Spain).
- Accelerated serving latency to **84.2 ms p50 (2.15x speedup)** at 17.3 QPS via ONNX Runtime FP16 compilation, alongside sub-50 ms in-browser edge screening, PostgreSQL RLS, and **253 passing automated tests (100% pass rate)**.
- **Research Manuscript (In Preparation):** Lead author of comprehensive biomedical informatics paper documenting TC-MBE ensemble, OAC-DG guardrails, US-CRC conformal risk control, and PASG-GradCAM saliency grounding.

C.O.P.P.E.R. -- Local AI Operating System & Microservices Engine | *FastAPI, Electron, React 19, Python, Docker* GitHub

- Architected a 30-agent orchestration hierarchy with a topological cascade router dispatching tasks at **~9,856 QPS** (<0.10 ms cache dispatch) across 1,390 benchmark test cases, validated against **538 automated tests**.
- Constructed Forge Sandbox with AST security validation and 4-tier friction protocol, achieving **100% threat sensitivity across 350 adversarial triggers** and Zero-Trust credential firewall.

Technical Skills

AI/ML & Computer Vision: PyTorch, OpenCV, Deep Vision (DenseNet, ConvNeXt, EfficientNet), Grad-CAM (XAI), Medical Image Processing, Model Calibration (Platt / ECE), Conformal Prediction, scikit-learn

Edge AI, HPC & Serving: ONNX Runtime (FP16 Graph Optimization), TensorRT, Distributed Model Serving, Redis, Celery Worker Queues, Asyncio Concurrency

Speech & Vernacular NLP: Speech-to-Text (Whisper), Bengali Audio Transcription, Local LLMs (Ollama), LangChain, Vector DBs (pgvector, ChromaDB)

Languages & Systems: Python, C++, Java, SQL (PostgreSQL RLS, SQLite WAL), JavaScript, TypeScript, FastAPI, React 19, Docker, Linux/Unix CLI, Git, CI/CD

Certifications

- **Azure AI Developer Associate (AI-103)** | Microsoft
- **Machine Learning Operations Engineer (AI-300)** | Microsoft
- **Oracle AI Database Foundations (1Z0-1195)** | Oracle
- **Oracle Agentic AI Foundations (1Z0-1157)** | Oracle
- **AI-ML Virtual Internship Accreditation** | All India Council for Technical Education (AICTE)